

How the Copilot Decides: Approve, Manual Review, or Decline

A quick reference for how every case lands in a bucket. — Underwriting Copilot

1. The score

Every applicant gets one **Composite Risk Score (0–100)**:

$$\text{Composite Score} = 50\% \times \text{Rule Engine score} + 50\% \times \text{ML model score}$$

- **Rule engine** — a fully auditable, hand-documented checklist (age, tobacco, BMI, medical conditions, credit, driving record, etc.). Every point traces to a written weight.
- **ML model** — a gradient-boosting model trained on historical outcomes. It catches interactions between factors the rules can't (e.g. "smoker + high BMI + young age" behaving differently than any one factor alone).

Blending them means one bad model can never single-handedly approve a risky case, and the rules alone can never miss a pattern the data has learned.

2. The score sets the band — nothing else

SCORE	BAND	OUTCOME
0–50	 Green	APPROVE — straight-through, no human touch
51–89	 Yellow	MANUAL REVIEW — a human underwriter decides
90–100	 Red	DECLINE — risk exceeds appetite

The band is **score-driven only**. A 22 is always green; a 95 is always red — regardless of anything else on the file. This is deliberate: it keeps the line defensible and auditable ("why was this declined?" always has the same one-line answer — the score).

3. The one exception: material misrepresentation

If the application **contradicts the medical or identity evidence** (e.g. the applicant said non-smoker but the lab came back positive, or the DOB doesn't match records), the case is **declined regardless of score**. This isn't a risk judgment — it's fraud/integrity, and it overrides everything else.

4. Flags: informative, not decisive

These show up on the case as **flags for the reviewer** — they explain context, but they do **not** move the case between bands:

- A major cross-document conflict (non-fraud)
- The affordability screen failing (can't justify the coverage on the applicant's income)
- The applicant disclosing unique circumstances

- The rule engine and ML model disagreeing sharply

A 40-score case with an affordability flag is still **green/APPROVE** — the flag just tells the underwriter (or whoever opens the file later) that something's worth a second look, without silently re-routing a clean-scoring case into someone's queue.

5. Who sees what

ROLE	VIEW
Underwriter	Works the manual-review queue only — the yellow-band cases, ranked by coverage + time waiting, not by score
Manager	Portfolio & Model Card — the score bands, formula, and full case list
Executive	Aggregate money view — coverage accepted/declined, approval rate, appetite tracking. No individual cases
Admin	Every recorded decision + outstanding evidence requests, for audit/compliance

6. Why underwriters don't rank by risk score

The manual-review queue is sorted by **coverage size + how long it's been waiting**, not by score. Sorting by score would mean the model's opinion decides who gets looked at first — which defeats the point of human oversight. A \$1M case sitting for 16 hours jumps the queue ahead of a \$50k case that just arrived, even if the \$50k case scores higher.